



High pressure densities and derived thermodynamic properties of deep eutectic solvents with menthol and saturated fatty acids

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ABSTRACT

Densities of four Deep Eutectic Solvents (DESs) based on menthol and organic acids (menthol:octanoic acid (1:1), menthol:decanoic acid (1:1), menthol:dodecanoic acid (2:1) and decanoic acid:octanoic acid (1:3)) were measured at high pressures (0.1 to 60) MPa and several temperatures (293.15 to 413.15) K. Tammann-Tait equation was used to model the experimental density data; the average absolute percentage deviations between calculated and experimental data were less or equal 0.012 %. The influence of the temperature and pressure on the derived volumetric properties: isothermal compressibility (κ_T), isobaric thermal expansivity (α_p), internal pressure (p_{int}) and difference between heat capacity at constant pressure (c_p) and constant volume (c_v) was analysed. Considering the experimental results, it is possible to observe that deep eutectic solvent formed by linear organic acids presents higher density values than those formed by menthol, being the densest the solvent formed by dodecanoic and octanoic acid. Attending to the derived volumetric properties in general, isothermal compressibility and isobaric thermal expansivity increase as temperature increases and decrease with pressure for all studied systems, being the octanoic acid based DESs slightly more compressible than the other studied DESs.

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1. Introduction

Volatile organic compounds (VOCs) are used in many industries as common solvents to remove pollutant substances from water or to recover valuable products obtained through industrial activities. Nevertheless, these solvents are toxic, volatile and harmful for human health and aquatic life, so it is necessary to replace them by other less hazardous compounds [1–3]. Over the last decades ionic liquids (ILs) generated great interest as possible substitution substances of VOCs, due to their unique properties such as high thermal stability, low vapour pressure and low melting points, or the possibility to design their dissolving properties [4–6]. However, ILs show some disadvantages such as their high price as well as the complex procedure of the synthesis [7,8].

Deep eutectic solvents (DESs) are easier to prepare and they present a lower cost than the ILs due to the lower price of their starting compounds. They are obtained from natural or biodegradable compounds, being in general less toxic than ILs. Similar to ILs,

DESs can be designed to have hydrophobic or hydrophilic properties and to dissolve organic or inorganic substances, therefore DES can be a good replacement for ILs [8,9].

DESs are prepared by mixing hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA) in an established molar ratio through the formation of hydrogen bonds, resulting in a mixture with a different and lower melting point than the starting compounds [10,11].

The knowledge of thermophysical properties of solvents is of the highest importance to assess and design numerous industrial processes or to develop the state equations with theoretical calculations. For this purpose, density and the derived volumetric properties are some of the most important data [4,12]. This information together with the influence of temperature on these properties help us to understand their behaviour and to find new and better applications of these solvents [13,14].

Nowadays, there are few articles about the physical properties of DESs, most of them are based on choline chloride as HBA with different HBD. Densities of mixtures prepared with water and DES formed with choline chloride and urea were measured at several temperatures by Yadav et al. [15]. Harifi-Mood et al. [16] determined density and viscosity for choline chloride and ethylene

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glycol based DES and excess molar volume for its mixtures with DMSO. In other work, DESs formed with choline chloride and levulinic acid, ethylene glycol or phenol and their mixtures with 1-butanol were studied by Gajardo-Parra et al. [1]. However, as far as the authors know, only Leron et al., [9,17,18] and recently Crespo et al. [19] reported physical properties of DESs at high pressures. Densities of choline chloride-urea and its aqueous mixtures were determined from 298.15 to 323.15 K at up to 50 MPa by Leron et al. [9]. In the paper of Leron and Li [17] volumetric properties of choline chloride-ethylene glycol and one mixture with water in established molar ratio were presented while the same authors measured solubility of carbon dioxide in choline chloride-glycerol at moderate pressures [18]. Crespo et al. [19] also examined DESs of cholinium chloride with ethylene glycol, glycerol or urea but at somewhat wider range of temperature and at pressures up to 100 MPa.

Similar to choline chloride based DESs, menthol and organic acid based DESs are also biodegradable and non-toxic, but their physical properties, such as viscosity, are lower than those of choline chloride based DESs, which makes them more applicable as extraction solvents [20,21].

In the last few years, different studies about menthol, thymol and organic acids based DESs were carried out. Their application as extracting solvents for different substances, such as alcohols or phenolic compounds (e.g. bisphenol A), as well as their physical properties (density, viscosity, vapour pressure or surface tension), have been studied by different authors. The mutual solubility of these hydrophobic solvents with water is also an interesting information for their potential use as extraction agents in aqueous solutions. Furthermore, the water content of DESs has an important influence on their physical properties [10,22–25].

The main objective of this work is to contribute to the knowledge on physical properties of these promising solvents. Therefore, densities of four DESs, menthol:octanoic acid (1:1) [Mth:Oct], menthol:decanoic acid (1:1) [Mth:Dec], menthol:dodecanoic acid (2:1) [Mth:Dod] and dodecanoic acid:octanoic acid (1:3) [Dod:Oct], were measured at several temperatures from 293.15 to 413.15 K and at pressures from 0.1 to 60 MPa. Moreover, the density data were correlated using the Tammann-Tait equation, which enabled the calculation of the isothermal compressibility (k_T), the isobaric thermal expansivity (α_p), the internal pressure (p_{int}) and the difference between specific heat capacities at constant pressure and constant volume ($c_p - c_v$).

2. Experimental section

2.1. Materials

The four studied DESs were prepared by mixing the pure compounds DL-menthol (designated as menthol in the text), octanoic acid, decanoic acid and dodecanoic acid in the established mole ratios: menthol:octanoic acid (1.0:1.0), menthol:decanoic acid (1.0:1.0), menthol:dodecanoic acid (2.0:1.0) and dodecanoic acid:octanoic acid (1.0:3.0) [10,22]. The chemical structures of the reagents are presented in Fig. 1 while Table 1 provides the information about the pure compounds used for the preparation of the studied DESs (Table 2).

2.2. Apparatus

DESs were prepared by weighing using a Mettler Toledo Excelent plus XP205 analytical balance, with a standard uncertainty of $\pm 3 \cdot 10^{-4}$ g and they were used without further purification. The standard uncertainty estimated for mole ratios of the components in prepared DES was ± 0.1 . ^1H NMR Bruker DPX 400 equipment was used to analyse the obtained DESs, the samples were prepared

using CDCl_3 as solvent and measured at 400 MHz. A Mettler Toledo C20 Coulometric KF Titrator was used to quantify the water content in the studied DESs as well as to check hydrophobicity of these solvents, with a relative standard deviation of 0.25%.

DMA HP measuring cell, from Anton Paar, was used for density (ρ) measurements at high temperatures and pressures. It was connected to DMA 5000 densimeter serving as a reading device [26]. The measurements temperature (T) was controlled by Peltier thermostat integrated in DMA HP with the expanded uncertainty, confidence level 95 % (coverage factor $k = 2$), of $U(T) = 0.1$ K. The pressure (p) control was achieved by means of pressure generator, High Pressure Equipment Co. model no. 50-6-15, with acetone as pressurizing fluid. Pressure transducer WIKA S-10 was used for pressure detection and the estimated expanded uncertainty, confidence level 95 % ($k = 2$), for pressure was $U(p) = 0.1$ MPa.

The main issue of working with DMA HP is that it does not give measured densities directly but an oscillation period of the measuring cell which depends on mass and, therefore, on density of the inserted sample. Hence, it is necessary to choose adequate calibration procedure to calculate densities from the obtained periods of oscillation. The classical calibration method with one reference fluid was applied in this study [27,28]. The detailed description of the used apparatus and the calibration procedure were described in other published work [26].

Knowing that viscosity affects density measurements using devices with oscillating element, its influence was quantified with empirical equation proposed by producer Anton Paar for DMA HP device. The viscosities of the studied DESs were taken from the literature [11,29] and density correction due to viscosity was estimated to less than $0.1 \text{ kg} \cdot \text{m}^{-3}$ for [Mth:Oct], [Mth:Dec], and [Dod:Oct]. Since menthol:dodecanoic acid (2:1) has significantly higher viscosity [11] compared to the other studied DESs, and even solidifies at lower temperatures, the density correction due to the damping effect for this sample was estimated to $0.4 \text{ kg} \cdot \text{m}^{-3}$.

The overall combined uncertainty of measured densities was calculated considering the uncertainties of the temperature, pressure and oscillation period measurements, the uncertainty reported for density of reference fluid used in calibration procedure, as well as the damping effect on the vibrating tube. The expanded combined uncertainty of the performed density measurements for all studied deep eutectic solvents, confidence level of 95% ($k = 2$), was estimated to $U(\rho) = 0.9 \text{ kg} \cdot \text{m}^{-3}$ at temperatures (293.15–363.15) K and $U(\rho) = 1.3 \text{ kg} \cdot \text{m}^{-3}$ at 373.15 K and higher temperatures.

3. Results and discussion

3.1. DESs formation

With the aim of obtaining the deep eutectic solvents, the starting compounds, in an appropriate proportion, were mixed, heated and stirred at moderate temperature (343.15 K) and at atmospheric pressure until getting a homogenous liquid. Then the mixtures were kept at room temperature to confirm their stability during 48 h.

The prepared solvents were analysed by ^1H NMR technique (Figs. S1 and S2) and the performed analyses showed that no reaction took place between components in the produced DESs. The NMR spectra are also significant for the confirmation of the unchanged structure of DESs recovered from various processes prior to their re-use [30].

3.2. Density measurements

Densities of the studied DESs were measured at temperatures (293.15–413.15) K and pressures (0.1–60) MPa and the obtained results are presented in Table 3 and plotted in Fig. S3.

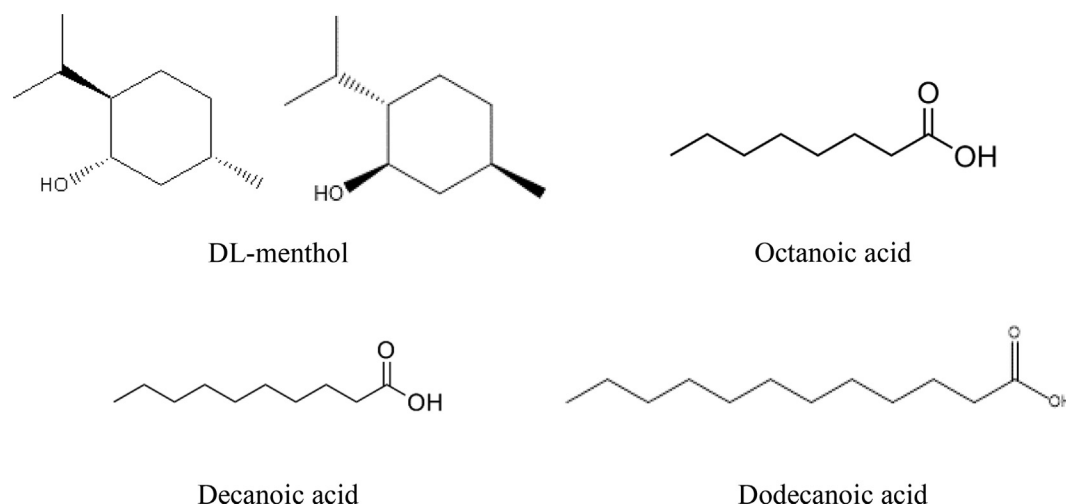


Fig. 1. Chemical structure of the reagents used in this work.

Table 1

Supplier, CAS number and mass fraction purity of the reagents used to obtain DES solvents.

Compound	Supplier	CAS #	Mass fraction purity ^a	Molecular weight (g·mol ⁻¹)
DL-menthol	Sigma-Aldrich	89-78-1	≥ 0.98	156.27
Octanoic acid (Caprylic acid)	Sigma-Aldrich	124-07-2	≥ 0.98	144.21
Decanoic acid (Capric acid)	Sigma-Aldrich	334-48-5	≥ 0.98	172.26
Dodecanoic acid (Lauric acid)	Scharlab	143-07-7	≥ 0.99	200.32

^a Provided by the suppliers.

Table 2

Molar ratio, molecular weight and water content of the produced liquid DESs at 293.15 K and 0.1 MPa.

DES	Molar ratio ^a	Molecular weight (g·mol ⁻¹)	Water content (ppm) ^b
Menthol:Octanoic acid	1.0:1.0	150.24	1908
Menthol:Decanoic acid	1.0:1.0	164.26	1704
Menthol:Dodecanoic acid	2.0:1.0	170.95	2315
Dodecanoic acid:Octanoic acid	1.0:3.0	158.24	149

Standard uncertainties:

^au (molar ratio) = ±0.1;^bu (water content) = ±0.25%;

u (T) = ±0.1 K; u (p) = ±0.001 MPa.

Since no published density data at higher pressures were found, densities obtained at atmospheric pressure by means of DMA HP measuring cell (Table 3) were compared to literature values [10,11,23,31] (Fig. 2). As for [Mth:Oct], average absolute percentage deviations (AAD) of experimentally determined densities from the literature [31] were less than 0.02 % (about 0.15 kg·m⁻³) and 0.36 % (about 3.15 kg·m⁻³) from the results of Nunes et al. [24]. Data measured for [Mth:Dec] agree good with literature data [31] with AAD less than 0.13 %. Measured densities of [Mth:Dod] match literature data [11] very well, AAD = 0.09 % (about 0.66 kg·m⁻³) and even better those of Verma and Banerjee [23], AAD = 0.04 % (about 0.33 kg·m⁻³). In the case of [Dod:Oct], excellent agreement between the experimental and published data was observed, AAD = 0.002 % (about 0.02 kg·m⁻³) for [31] and 0.05 % (about 0.46 kg·m⁻³) for [10].

It is evident that the agreement of experimental data for [Mth:Oct] with the values published by Nunes et al. [24] (Fig. 2a) is not very good, especially at lower temperatures. Having in mind that density data given in [24] do not show linear temperature dependence, as it is expected for density, the reliability

of data reported by Nunes et al. [24] seems questionable. On the other hand, the mentioned experimental densities agree very well with the results of Sas et al. [31]. Very good agreement of the densities measured at atmospheric pressure with the literature data (Fig. 2) proves the reliability of the applied measurement and calibration procedures and the accuracy of the data presented in this paper.

Data given in Fig. 3 represent densities of the studied DESs obtained at boundary values of examined temperature interval, while all measured densities are shown in Fig. S3 in the supplementary material to this paper. As expected, density increases with an increase in pressure and decreases when temperature rises. The influence of temperature on density is greater than of a pressure.

It can be concluded that the studied deep eutectic solvents made from octanoic acid have higher densities than the other two (Table 3, Fig. 3). Analyse of DESs with menthol reveals that [Mth:Oct] has the highest and [Mth:Dod] the lowest density which indicates that the use of carboxylic acids with longer chain for DES production leads to its lower density. Density comparison for DESs with dodecanoic acid shows that [Mth:Dod] is less dense than [Dod:Oct], i.e. combination of two carboxylic acids has denser packaging than strait chain carboxylic acid with cyclic alcohol. It can be due to the fact that solvents are obtained by the formation of hydrogen bonds established between hydroxyl groups present in the starting compound structures (organic acids and menthol). However, in the solvent composed by linear acids the formation of van der Waals interactions between alkyl chains and better geometrical fitting of linear chains is possible which produces better packing and therefore denser solvents. On the other hand, the studied DESs produced from octanoic acid ([Mth:Oct] and [Dod:Oct]) had almost the same density. This may indicate that in the case of the carboxylic acids with shorter chain (such as octanoic acid) the similar packaging will be achieved with either the cyclic alcohol (menthol) or linear carboxylic acid with longer chain (dodecanoic acid).

Table 3

Densities of [Mth:Oct] (molar ratio: 1.0:1.0), [Mth:Dec] (1.0:1.0), [Mth:Dod] (2.0:1.0) and [Dod:Oct] (1.0:3.0) in liquid phase at temperature T and pressure p.

p ^c /MPa	$\rho^a/\text{kg}\cdot\text{m}^{-3}$										
	T ^b /K										
	293.15	303.15	313.15	323.15	333.15	343.15	353.15	363.15	373.15	393.15	413.15
Menthol:octanoic acid (molar ratio: 1.0:1.0) ^d											
0.1	905.6	898.0	890.5	882.8	875.1	867.3	859.5	851.5	842.8	826.4	810.1
1	906.1	898.6	891.1	883.5	875.8	868.0	860.2	852.3	843.9	827.5	811.0
5	908.6	901.2	893.7	886.3	878.7	871.1	863.4	855.7	847.5	831.5	815.4
10	911.6	904.3	897.0	889.7	882.3	874.9	867.4	859.8	851.8	836.3	820.7
15	914.5	907.3	900.2	893.0	885.8	878.5	871.2	863.8	856.0	840.9	825.8
20	917.3	910.3	903.3	896.2	889.1	882.0	874.8	867.6	860.0	845.3	830.7
25	920.1	913.2	906.2	899.3	892.4	885.4	878.4	871.3	863.8	849.6	835.3
30	922.7	915.9	909.1	902.3	895.5	888.6	881.8	874.9	867.6	853.6	839.8
35	925.3	918.6	911.9	905.2	898.5	891.8	885.1	878.3	871.1	857.5	844.0
40	927.9	921.2	914.7	908.0	901.4	894.8	888.2	881.5	874.5	861.2	848.0
45	930.3	923.8	917.3	910.8	904.3	897.7	891.3	884.7	877.8	864.7	851.7
50	932.7	926.2	919.8	913.4	907.0	900.5	894.1	887.7	880.8	868.0	855.3
55	935.0	928.6	922.2	915.9	909.6	903.2	896.9	890.5	883.8	871.1	858.6
60	937.2	930.8	924.6	918.3	912.0	905.8	899.5	893.2	886.6	874.0	861.8
Menthol:decanoic acid (molar ratio: 1.0:1.0) ^d											
0.1	900.3	892.9	885.5	878.0	870.3	862.7	855.0	847.5	839.2	823.6	807.4
1	900.9	893.5	886.1	878.6	871.0	863.4	855.7	848.3	840.2	824.7	808.3
5	903.2	896.0	888.7	881.4	873.8	866.4	858.9	851.6	843.7	828.5	812.6
10	906.2	899.0	891.9	884.7	877.3	870.0	862.7	855.6	847.9	833.2	817.7
15	909.0	902.0	895.0	887.9	880.7	873.4	866.4	859.5	852.0	837.7	822.7
20	911.8	904.8	898.0	891.1	883.9	876.8	870.0	863.2	855.9	842.0	827.4
25	914.5	907.6	900.9	894.1	887.1	880.1	873.4	866.8	859.7	846.1	831.9
30	917.1	910.3	903.7	897.1	890.2	883.3	876.7	870.3	863.3	850.0	836.2
35	919.6	912.9	906.5	899.9	893.1	886.4	879.9	873.6	866.7	853.8	840.3
40	922.1	915.5	909.1	902.6	896.0	889.4	883.0	876.8	870.1	857.4	844.2
45	924.5	918.0	911.7	905.3	898.7	892.3	886.0	879.9	873.2	860.8	847.9
50	926.8	920.4	914.2	907.9	901.3	895.1	888.8	882.8	876.2	864.0	851.3
55	929.1	922.7	916.6	910.3	903.9	897.7	891.5	885.6	879.1	867.1	854.6
60	931.2	924.9	918.9	912.7	906.3	900.3	894.1	888.2	881.9	870.0	857.6
Menthol:dodecanoic acid (molar ratio: 2.0:1.0) ^{d,e}											
0.1	889.6	882.3	874.9	867.5	859.9	852.2	844.6	836.5	820.9	805.1	
1	890.2	882.9	875.5	868.2	860.6	852.9	845.3	837.6	821.9	806.0	
5	892.6	885.5	878.3	871.0	863.5	856.1	848.6	841.1	825.8	810.2	
10	895.6	888.6	881.5	874.4	867.1	859.8	852.6	845.2	830.4	815.4	
15	898.6	891.7	884.7	877.8	870.6	863.5	856.4	849.2	834.8	820.3	
20	901.4	894.6	887.8	881.0	873.9	867.0	860.0	853.0	839.0	824.9	
25	904.1	897.4	890.7	884.0	877.1	870.3	863.5	856.7	843.0	829.3	
30	906.8	900.2	893.6	887.0	880.2	873.5	866.9	860.2	846.8	833.4	
35	909.3	902.8	896.3	889.8	883.2	876.6	870.1	863.5	850.4	837.3	
40	911.8	905.3	898.9	892.6	886.0	879.5	873.1	866.7	853.8	840.9	
45	914.1	907.8	901.4	895.2	888.7	882.3	876.0	869.7	856.9	844.3	
Dodecanoic acid:octanoic acid (molar ratio: 1.0:3.0) ^d											
0.1	905.0	897.1	889.3	881.4	873.5	865.7	857.8	850.0	841.8	826.0	810.2
1	905.6	897.7	889.9	882.0	874.2	866.4	858.6	850.8	842.9	827.1	811.2
5	908.1	900.4	892.7	885.0	877.3	869.6	862.0	854.3	846.6	831.2	815.7
10	911.3	903.7	896.1	888.6	881.1	873.6	866.1	858.6	851.2	836.2	821.1
15	914.3	906.9	899.4	892.0	884.7	877.4	870.1	862.8	855.5	841.0	826.3
20	917.3	909.9	902.6	895.4	888.2	881.0	873.9	866.8	859.7	845.5	831.3
25	920.1	912.9	905.8	898.6	891.6	884.6	877.6	870.7	863.7	849.9	836.1
30	922.9	915.9	908.8	901.8	894.9	888.0	881.2	874.4	867.6	854.1	840.6
35	925.7	918.7	911.7	904.9	898.1	891.3	884.6	877.9	871.3	858.1	845.0
40	928.3	921.4	914.6	907.8	901.1	894.5	887.9	881.3	874.8	861.9	849.1
45	930.8	924.0	917.3	910.6	904.1	897.5	891.0	884.6	878.2	865.6	853.0
50	933.3	926.6	920.0	913.4	906.9	900.4	894.1	887.7	881.4	869.0	856.6
55	935.7	929.1	922.5	916.0	909.6	903.3	897.0	890.7	884.5	872.2	860.1
60	938.0	931.5	925.0	918.5	912.2	905.9	899.7	893.5	887.4	875.3	863.3

^{a-d} Expanded uncertainties with 95% level of confidence, U:^a U(ρ) = 0.9 kg·m⁻³ (293.15 K ≤ T ≤ 363.15 K) and 1.3 kg·m⁻³ (373.15 K ≤ T ≤ 413.15 K);^b U(T) = 0.1 K;^c U(p) = 0.1 MPa;^d U(molar ratio) = 0.2;^e The sample solidifies at lower temperatures and high pressures.

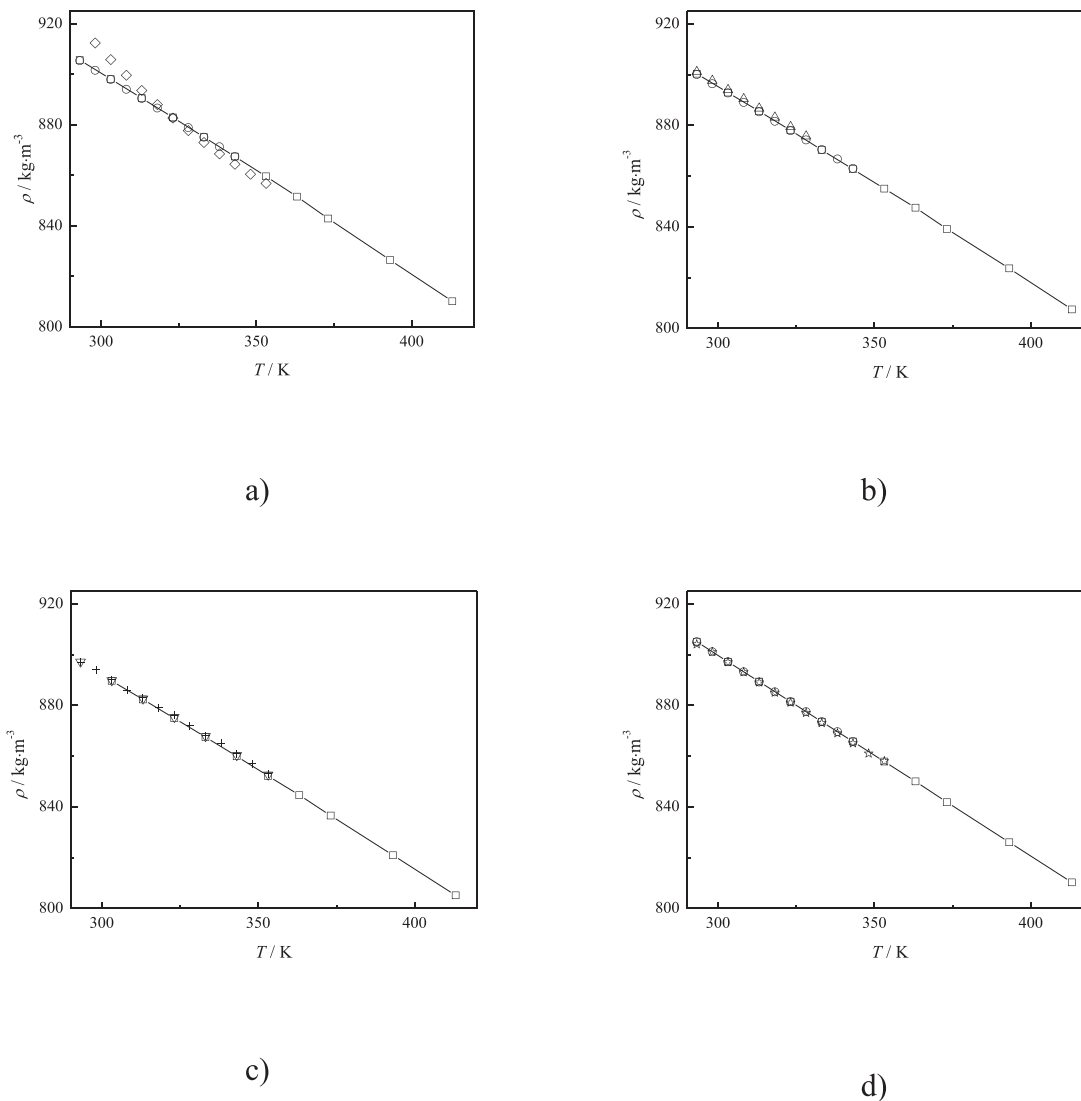


Fig. 2. Comparison of experimental density (ρ) data at atmospheric pressure ($\square$) for DES a) [Mth:Oct] (1.0:1.0), b) [Mth:Dec] (1.0:1.0), c) [Mth:Dod] (2.0:1.0) and d) [Dod:Oct] (1.0:3.0) with literature: ($\circ$) Sas et al. [31], ($\diamond$) Nunes et al. [24], ($\triangle$) Dietz et al. [25], (∇) Verma and Banerjee [23], (+) Ribeiro et al. [11] and ($\star$) Florindo et al. [10]. Lines are guides for the eye.

3.3. Density modelling

The measured densities of the studied deep eutectic solvents were fitted to the modified Tammann-Tait equation [32]:

$$\rho(T, p) = \frac{\rho^{\text{ref}}(T)}{1 - C \cdot \ln\left(\frac{B(T)+p}{B(T)+p^{\text{ref}}}\right)} \quad (1)$$

where ρ^{ref} stands for density of the sample at reference pressure (p^{ref}). Parameter $B(T)$ depends on temperature in a similar way as ρ^{ref} whereas C is the temperature independent parameter:

$$\rho^{\text{ref}}(T) = \sum_{i=0}^2 a_i T^i \quad (2)$$

$$B(T) = \sum_{i=0}^2 b_i T^i \quad (3)$$

The density modelling started with the optimization of a_i parameters from the eq. (2) using densities measured at reference pressure $p^{\text{ref}} = 0.1$ MPa. The other adjustable parameters b_i and C

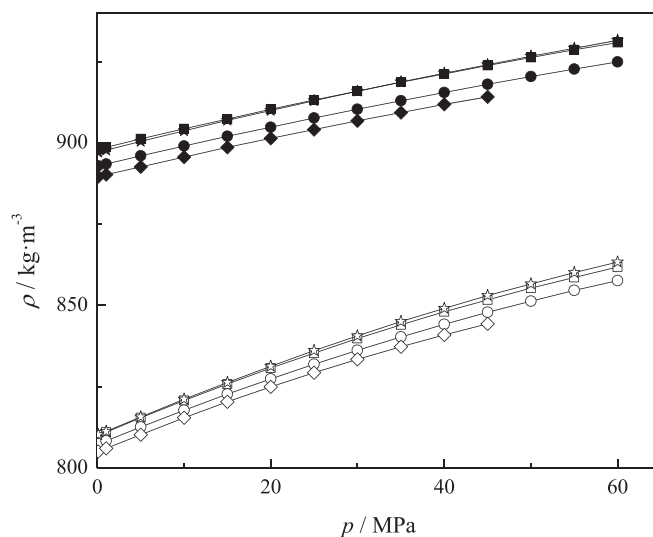


Fig. 3. Densities (ρ) as a function of pressure (p) of ($\blacksquare, \square$) [Mth:Oct] (1.0:1.0), ($\bullet, \circ$) [Mth:Dec] (1.0:1.0), ($\blacklozenge, \lozenge$) [Mth:Dod] (2.0:1.0) and ($\star, \star$) [Dod:Oct] (1.0:3.0) at 303.15 K (full symbols) and 413.15 K (empty symbols). Lines are guides for the eye.

Table 4

The Parameters of the Modified Tammann-Tait Equation.

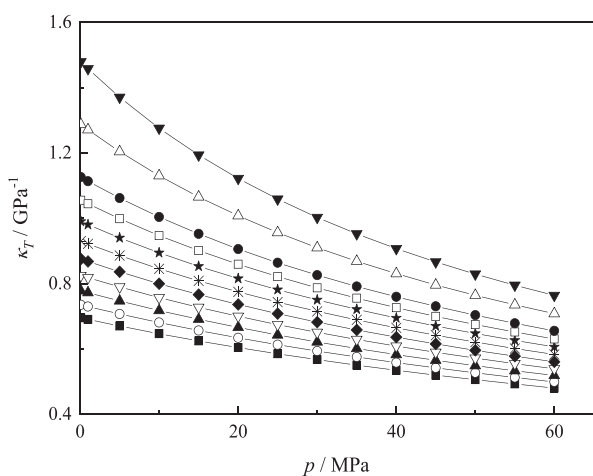
	[Mth:Oct] (molar ratio: 1.0:1.0)	[Mth:Dec] (molar ratio: 1.0:1.0)	[Mth:Dod] (molar ratio: 2.0:1.0)	[Dod:Oct] (molar ratio: 1.0:3.0)
$a_0/\text{kg}\cdot\text{m}^{-3}$	1087.08	1088.34	1079.93	1131.31
$a_1/\text{kg}\cdot\text{m}^{-3}\cdot\text{K}^{-1}$	-0.49231	-0.54733	-0.52323	-0.75944
$a_2/\text{kg}\cdot\text{m}^{-3}\cdot\text{K}^{-2}$	$-4.3246\cdot 10^{-4}$	$-3.2081\cdot 10^{-4}$	$-3.4435\cdot 10^{-4}$	$-4.3070\cdot 10^{-5}$
b_0/MPa	409.183	426.282	390.389	435.784
$b_1/\text{MPa}\cdot\text{K}^{-1}$	-1.3094	-1.3822	-1.2483	-1.4854
$b_2/\text{MPa}\cdot\text{K}^{-2}$	$1.1021\cdot 10^{-3}$	$1.1930\cdot 10^{-3}$	$1.0529\cdot 10^{-3}$	$1.3777\cdot 10^{-3}$
C	$8.3458\cdot 10^{-2}$	$8.3943\cdot 10^{-2}$	$7.7655\cdot 10^{-2}$	$8.6324\cdot 10^{-2}$
AAD ^a /%	0.012	0.012	0.006	0.008
MD ^b /%	0.047	0.046	0.023	0.044
Bias ^c /%	0.001	0.001	0.0003	0.001
$\sigma^d/\text{kg}\cdot\text{m}^{-3}$	0.141	0.145	0.067	0.101

$$^a\sigma = \sqrt{\frac{\sum_{i=1}^N (\rho_i^{\text{exp}} - \rho_i^{\text{cal}})^2}{N-m}}$$

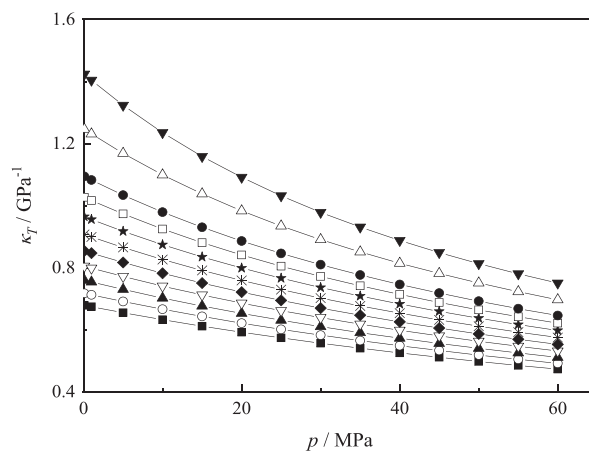
$$^b\text{AAD} = \frac{100}{N} \sum_{i=1}^N \left| \frac{\rho_i^{\text{exp}} - \rho_i^{\text{cal}}}{\rho_i^{\text{exp}}} \right|$$

$$^c\text{MD} = \max \left(100 \left| \frac{\rho_i^{\text{exp}} - \rho_i^{\text{cal}}}{\rho_i^{\text{exp}}} \right| \right), i = 1, N.$$

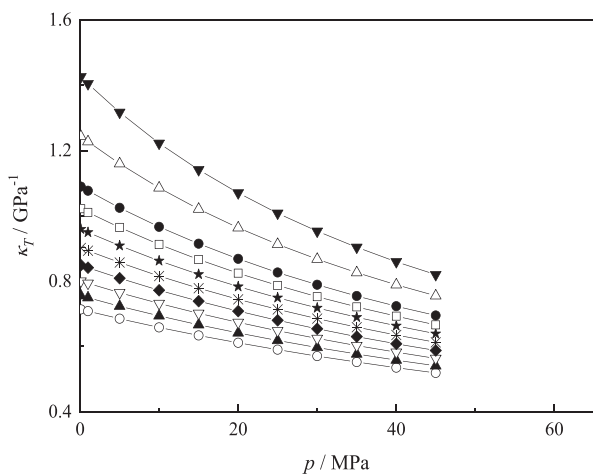
$$^d\text{Bias} = \frac{100}{N} \sum_{i=1}^N \frac{\rho_i^{\text{exp}} - \rho_i^{\text{cal}}}{\rho_i^{\text{exp}}}$$



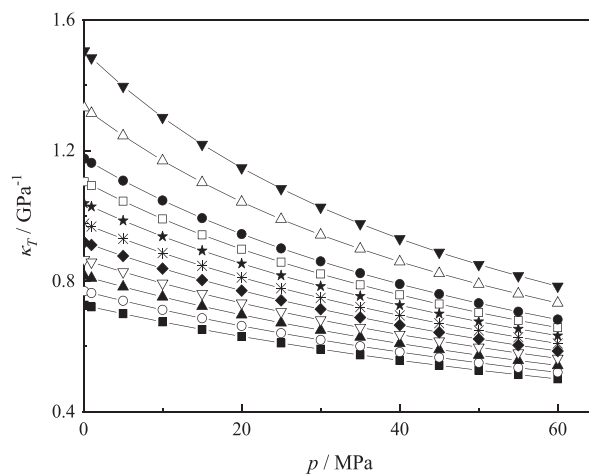
a)



b)



c)



d)

Fig. 4. The isothermal compressibilities (κ_T) as a function of pressure (p) of: a) [Mth:Oct] (1.0:1.0), b) [Mth:Dec] (1.0:1.0), c) [Mth:Dod] (2.0:1.0) and d) [Dod:Oct] (1.0:3.0) at (■) 293.15 K, (○) 303.15 K, (▲) 313.15 K, (▽) 323.15 K, (◆) 333.15 K, (*) 343.15 K, (★) 353.15 K, (□) 363.15 K, (●) 373.15 K, (△) 393.15 K and (▼) 413.15 K. Lines are guides for the eye.

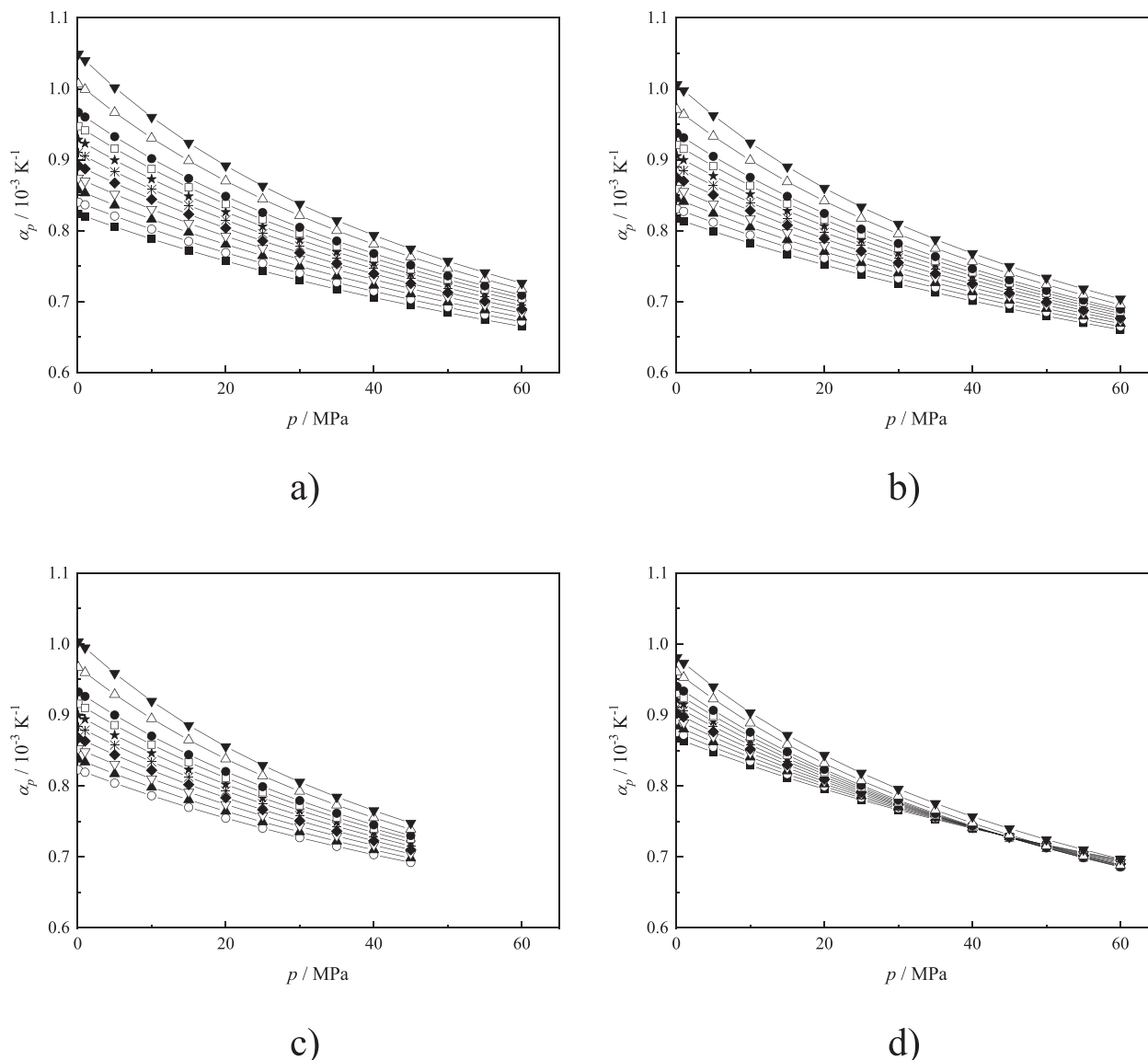


Fig. 5. The isobaric thermal expansivities (α_p) as a function of pressure (p) of: a) [Mth:Oct] (1.0:1.0), b) [Mth:Dec] (1.0:1.0), c) [Mth:Dot] (2.0:1.0) and d) [Dot:Oct] (1.0:3.0) at (■) 293.15 K, (○) 303.15 K, (▲) 313.15 K, (▽) 323.15 K, (◆) 333.15 K, (✱) 343.15 K, (★) 353.15 K, (□) 363.15 K, (●) 373.15 K, (△) 393.15 K and (▼) 413.15 K. Lines are guides for the eye.

were optimized using the eq. (1), including determined parameters a_i , and the rest of the density data, at pressures higher than 0.1 MPa. The parameters were optimized by applying Levenberg–Marquardt algorithm [33] where the objective function was standard deviation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (\rho_i^{\text{exp}} - \rho_i^{\text{cal}})^2}{N - m}} \quad (4)$$

where, ρ^{exp} denotes the experimentally measured and ρ^{cal} the calculated density using the eq. (1), N is the number of experimental data while m refers to the number of the optimized parameters of the modified Tammann-Tait equation. The results of the high pressure density modelling, together with the optimization quality criteria [34], i.e. the absolute average percentage deviation (AAD), the percentage maximum deviation (MD), the average percentage deviation (Bias), and the standard deviation (σ), are presented in Table 4.

The quality criteria listed in Table 4 prove the accuracy of the obtained parameters of the modified Tammann-Tait equation.

3.4. Derived properties

Density modelling enables the determination of the derived thermodynamic properties which are very important both for the experimental research as well as for the industry.

The isothermal compressibility (κ_T) represents the density change due to change of pressure at constant temperature [34,35]:

$$\kappa_T = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \quad (5)$$

Similar to that, the isobaric thermal expansivity (α_p) shows the influence of temperature change on density at constant pressure [34,35]:

$$\alpha_p = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p \quad (6)$$

The internal pressure (p_{int}) describes the change of the internal energy as a response to the change of the volume of the sample when temperature is constant. This property is valuable for understanding the intermolecular interactions in the sample [34,35]:

$$p_{\text{int}} = \left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial p}{\partial T} \right)_p - p = T\gamma - p = \frac{T\alpha_p}{\kappa_T} - p \quad (7)$$

where U stands for an internal energy and V denotes volume of the sample.

Knowledge of abovementioned properties makes it easy to determine heat capacity at constant volume (c_v). Namely, measuring of isochoric heat capacity can be quite complex so it is significantly easier to calculate it if the heat capacity at constant pressure (c_p) is known.

$$c_p = c_v + T \frac{\left(\frac{\partial p}{\partial T} \right)_p^2}{\rho^2 \left(\frac{\partial p}{\partial \rho} \right)_T} \quad (8)$$

When the eqs. (5) and (6) are included in the eq (8) the term for the difference between the heat capacities at constant pressure and at constant volume is derived [34,35]:

$$c_p - c_v = \frac{\alpha_p^2 T}{\rho \kappa_T} \quad (9)$$

The calculated values for the isothermal compressibility, the isobaric thermal expansivity, the internal pressure and the difference between specific heat capacities at constant pressure and at constant volume for the examined deep eutectic solvents in the temperature interval (293.15–413.15) K and at pressures (0.1–60) MPa are given in the [supplementary material](#) to this paper (Tables S1–S4).

The isothermal compressibilities of all examined deep eutectic solvents show expected behaviour. They increase when temperature rises at constant pressure and decrease along isotherms with increase in pressure (Tables S1–S4, Fig. 4). The temperature influence is more visible at lower pressures close to the atmospheric while pressure affects the isothermal compressibility more at higher temperatures. It is noticeable that deep eutectic solvents containing octanoic acid are slightly more compressible than the other two studied samples (Fig. 4).

As expected, the isobaric thermal expansivity rises when temperature goes up along isobars and decreases when pressure increases at constant temperature (Tables S1–S4, Fig. 5). It is evident that the temperature influence on the isobaric thermal expansivity of the solvent formed by linear acids [Dod:Oct] is slightly weaker compared to the solvents containing menthol. The intersection point of α_p isotherms (pressure where α_p is temperature independent) occurs at 45 MPa for [Dod:Oct]. At pressures higher than 45 MPa the dependence of α_p on temperature becomes opposite, it decreases when temperature rises. The isotherms of the isobaric thermal expansivity for the deep eutectic solvents made from menthol approach each other when pressure rises but their intersection point can be expected above 60 MPa.

As for the difference between the isobaric and isochoric heat capacities, for all the examined DESs it was noted that the values calculated at atmospheric pressure were higher than those obtained at higher pressures. Opposite to that, difference between the isobaric and isochoric heat capacities increases with temperature rise. In the case of DES consisting of two linear carboxylic acids ([Dod:Oct]) the mentioned dependence of $c_p - c_v$ on temperature was observed only at temperatures up to 323.15 K and as temperature increases above that one the difference between the isobaric and isochoric heat capacities decreases.

As it was stated before, the internal pressure describes the change of the internal energy as a response to the change of the sample volume at constant temperature. Typically, when the attractive forces are dominant the internal pressure increases under expansion (with the pressure decrease). For the analysed DESs this happens at around room temperature (see Figs. 6 and

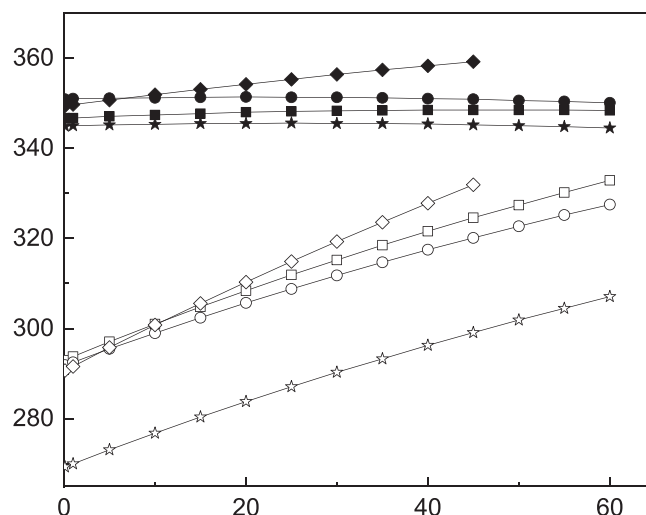


Fig. 6. Internal pressure (p_{int}) as a function of pressure (p) of (■, □) [Mth:Oct] (1.0:1.0), (●, ○) [Mth:Dec] (1.0:1.0), (◆, ◇) [Mth:Dod] (2.0:1.0) and (★, ☆) [Dod:Oct] (1.0:3.0) at 303.15 K (full symbols) and 413.15 K (empty symbols). Lines are guides for the eye.

S4), p_{int} decreases negligibly with increasing pressure. The opposite, increasing trend in p_{int} is evident for the [Mth:Dod], which has the lowest density comparing to the other three systems. This behaviour of p_{int} would implicate the absence of the intermolecular attractive forces, even at around room temperature. At higher temperatures distinctive increasing trend in p_{int} with pressure rise is obvious for all systems (Fig. 6). This behaviour of liquids at higher temperatures was described in the literature [36,37]. It is also evident that the internal pressure decreases with temperature rise for the same pressure conditions.

4. Conclusions

In this work density at high pressures and different temperatures was measured for four deep eutectic solvents based on fatty acids and menthol. The observed behaviour was as expected, the density of all studied solvents decreases as temperature increases and increases as pressure increases. The lowest values of density were obtained for [Mth:Dod] and octanoic acid based DESs present the highest density values. The experimental density data were fitted to the modified Tammann-Tait equation and the obtained deviations of the calculated from the experimental data indicated very good agreement between both of them.

Derived thermodynamic properties were calculated for the studied solvents. The isothermal compressibility (κ_T), the isobaric thermal expansivity (α_p), the internal pressure (p_{int}) and the difference between the heat capacities at constant pressure and at constant volume ($c_p - c_v$) were obtained from density values.

It was observed that the influence of temperature on isothermal compressibility is more noticeable than the influence of pressure. This property increases as temperature increases at constant pressure and decreases as pressure increases along isotherms. The octanoic acid based DESs are slightly more compressible than the other studied DESs.

Similar behaviour as for the isothermal compressibility was obtained for the isobaric thermal expansivity, i.e. increasing with temperature increase and decreasing with pressure at constant temperature. Nevertheless, the opposite behaviour was observed for [Dod:Oct] DES at pressures higher than 45 MPa, where an increase in temperature produces a decrease of this derived property.

The difference between the isobaric and isochoric heat capacities ($C_p - C_v$) decreases with pressure rise for all studied solvents. An increase in temperature enhances the increase of this property, except for [Dod:Oct] DES where the opposite behaviour was observed at temperatures higher than 323.15 K.

In general, the internal pressure decreases as temperature increases and as pressure decreases, especially at higher temperatures, implicating the low intensity intermolecular forces.

CRediT authorship contribution statement

Olalla G. Sas: Writing - original draft. **Gorica R. Ivaniš:** Writing - original draft. **Mirjana Lj. Kijevčanin:** Formal analysis, Funding acquisition, Writing - review & editing. **Begoña González:** Supervision, Validation, Writing - review & editing. **Angeles Domínguez:** Supervision, Writing - review & editing. **Ivona R. Radović:** Conceptualization, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jct.2021.106578>.

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